# Daily Literature Review — 2026-10-13

# Daily Literature Review: AI and machine learning in protein function prediction

Date: 2026-10-13

## Research Question

How are machine-learning methods being used to predict protein function, what datasets and evaluation metrics are used, and what limitations remain?

## Summary

This document compiles 40 recent papers related to the topic. AI drafting was unavailable, so the content below is built directly from bibliographic metadata. Each entry includes title, authors, year, source, DOI/URL, and a reminder to extract findings from the original paper. This is an evidence-collection scaffold, not a finished scientific review.

## Evidence Table (Bibliographic)

### 2026

**Rapid directed evolution guided by protein language models and epistatic interactions.**

- Authors: Tran VQ, Nemeth M, Bartie LJ, Chandrasekaran SS, Fanton A, Moon HC, Hie BL, Konermann S

- Source: PubMed

- PMID: 41712694

- Link: https://pubmed.ncbi.nlm.nih.gov/41712694/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Machine learning-driven drug repurposing for HER2-positive breast cancer.**

- Authors: Dinesh BGH, Ganjipete S, Kumar BS, Ammunje DN, Kunjiappan S, Seetharaman PK, Ramar M, Chidambaram K

- Source: PubMed

- PMID: 41942578

- Link: https://pubmed.ncbi.nlm.nih.gov/41942578/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Machine Learning for Studying Plant Protein Function and Evolution**

- Authors: Ankan Das, Dibyendu Seth, Bodhisattya Pal, Nidhi, Sandip Debnath

- Source: OpenAlex

- DOI: 10.1201/9781003545781-20

- Link: https://doi.org/10.1201/9781003545781-20

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Leveraging Gemma 4 Large Language Model for Protein Function Prediction and Interpretability Application of AI Models for Protein Function Prediction from Amino Acid Sequences**

- Authors: Trinh Quang Minh, Ngo Thi Lan

- Source: OpenAlex

- DOI: 10.38124/ijisrt/26apr247

- Link: https://doi.org/10.38124/ijisrt/26apr247

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

### 2025

**Predicting response and survival of lung adenocarcinoma under anti-programmed death-1 therapy using biological deep learning.**

- Authors: Wang Y, Zhang L, Xie H, Wang L, Wang Y, Li S, He J, Wang M

- Source: PubMed

- PMID: 40971820

- Link: https://pubmed.ncbi.nlm.nih.gov/40971820/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Real-world clinical multi-omics analyses reveal bifurcation of ER-independent and ER-dependent drug resistance to CDK4/6 inhibitors.**

- Authors: Kan Z, Wen J, Bonato V, Webster J, Yang W, Ivanov V, Kim KH, Roh W

- Source: PubMed

- PMID: 39843429

- Link: https://pubmed.ncbi.nlm.nih.gov/39843429/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Longitudinal single-cell RNA model aids prediction of EGFR-TKI resistance.**

- Authors: Hou G, Lu Z, Zeng D, Chen Q, Cheng S, Song B

- Source: PubMed

- PMID: 40739818

- Link: https://pubmed.ncbi.nlm.nih.gov/40739818/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Sequence-based virtual screening using transformers.**

- Authors: Zhang S, Huo D, Horne RI, Qi Y, Pujalte Ojeda S, Yan A, Vendruscolo M

- Source: PubMed

- PMID: 40721411

- Link: https://pubmed.ncbi.nlm.nih.gov/40721411/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**F-CPI: A Multimodal Deep Learning Approach for Predicting Compound Bioactivity Changes Induced by Fluorine Substitution.**

- Authors: Zhang Q, Yin W, Chen X, Zhou A, Zhang G, Zhao Z, Li Z, Zhang Y

- Source: PubMed

- PMID: 39707149

- Link: https://pubmed.ncbi.nlm.nih.gov/39707149/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**AI and Machine Learning in Biology: From Genes to Proteins**

- Authors: Zaw Myo Hein, Dhanyashri Guruparan, Blaire Okunsai, Che Mohd Nasril Che Mohd Nassir, Muhammad Danial Che Ramli, Suresh Kumar

- Source: OpenAlex

- DOI: 10.3390/biology14101453

- Link: https://doi.org/10.3390/biology14101453

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Modern machine learning methods for protein property prediction**

- Authors: Arjun Dosajh, Prakul Agrawal, Prathit Chatterjee, U. Deva Priyakumar

- Source: OpenAlex

- DOI: 10.1016/j.sbi.2025.102990

- Link: https://doi.org/10.1016/j.sbi.2025.102990

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**A REVIEW ON EXPLAINABLE AI IN ProTEIN FUNCTION**

- Authors: Aastha Katiyar

- Source: OpenAlex

- DOI: 10.22541/au.174912536.68729526/v1

- Link: https://doi.org/10.22541/au.174912536.68729526/v1

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**AI-Driven Functional Protein Design: Machine Learning Approaches for Screening and Optimization**

- Authors: Yunze Jiang

- Source: OpenAlex

- DOI: 10.1145/3778450.3778490

- Link: https://doi.org/10.1145/3778450.3778490

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

### 2024

**Machine learning-based reproducible prediction of type 2 diabetes subtypes.**

- Authors: Tanabe H, Sato M, Miyake A, Shimajiri Y, Ojima T, Narita A, Saito H, Tanaka K

- Source: PubMed

- PMID: 39168869

- Link: https://pubmed.ncbi.nlm.nih.gov/39168869/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Machine learning prediction of future amyloid beta positivity in amyloid-negative individuals.**

- Authors: Moradi E, Prakash M, Hall A, Solomon A, Strange B, Tohka J, Alzheimer’s Disease Neuroimaging Initiative

- Source: PubMed

- PMID: 38414035

- Link: https://pubmed.ncbi.nlm.nih.gov/38414035/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Construction of IRAK4 inhibitor activity prediction model based on machine learning.**

- Authors: Zhao Y, Wan Q, He X

- Source: PubMed

- PMID: 38970641

- Link: https://pubmed.ncbi.nlm.nih.gov/38970641/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Deep learning methods for protein function prediction**

- Authors: Frimpong Boadu, Ahhyun Lee, Jianlin Jack Cheng

- Source: OpenAlex

- DOI: 10.1002/pmic.202300471

- Link: https://doi.org/10.1002/pmic.202300471

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**AI-Driven Deep Learning Techniques in Protein Structure Prediction**

- Authors: Lingtao Chen, Qiaomu Li, Kazi Fahim Ahmad Nasif, Ying Xie, Bobin Deng, Shuteng Niu, Seyedamin Pouriyeh, Zhiyu Dai

- Source: OpenAlex

- DOI: 10.3390/ijms25158426

- Link: https://doi.org/10.3390/ijms25158426

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Machine Learning-Assisted Protein Structure Prediction: An AI Approach for Biochemical Insights**

- Authors: Radha Mahendran, Akriti Dogra, Dinesh Mendhe, S. B G Tilak Babu, Shriniket Dixit, Surya Pratap Singh

- Source: OpenAlex

- DOI: 10.1109/iconstem60960.2024.10568895

- Link: https://doi.org/10.1109/iconstem60960.2024.10568895

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Protein function prediction as approximate semantic entailment**

- Authors: Maxat Kulmanov, Francisco J. Guzmán■Vega, Paula D Duek, Lydie Lane, Stefan T. Arold, Robert Hoehndorf

- Source: OpenAlex

- DOI: 10.1038/s42256-024-00795-w

- Link: https://doi.org/10.1038/s42256-024-00795-w

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Improving protein function prediction by learning and integrating representations of protein sequences and function labels**

- Authors: Frimpong Boadu, Jianlin Jack Cheng

- Source: OpenAlex

- DOI: 10.1093/bioadv/vbae120

- Link: https://doi.org/10.1093/bioadv/vbae120

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Advances in AI and machine learning for predictive medicine**

- Authors: Alok K. Sharma, Artem Lysenko, Shangru Jia, Keith A. Boroevich, Tatsuhiko Tsunoda

- Source: OpenAlex

- DOI: 10.1038/s10038-024-01231-y

- Link: https://doi.org/10.1038/s10038-024-01231-y

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

### 2023

**Hierarchical graph learning for protein-protein interaction.**

- Authors: Gao Z, Jiang C, Zhang J, Jiang X, Li L, Zhao P, Yang H, Huang Y

- Source: PubMed

- PMID: 36841846

- Link: https://pubmed.ncbi.nlm.nih.gov/36841846/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Machine Learning-Guided Protein Engineering**

- Authors: Petr Kouba, Pavel Kohout, Faraneh Haddadi, Anton Bushuiev, Raman Samusevich, Ji■í Sedlá■, Ji■■■ Damborský, Tomáš Pluskal

- Source: OpenAlex

- DOI: 10.1021/acscatal.3c02743

- Link: https://doi.org/10.1021/acscatal.3c02743

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

### 2022

**Artificial intelligence based methods for hot spot prediction.**

- Authors: Ovek D, Abali Z, Zeylan ME, Keskin O, Gursoy A, Tuncbag N

- Source: PubMed

- PMID: 34954608

- Link: https://pubmed.ncbi.nlm.nih.gov/34954608/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Machine Learning Analysis of Cocaine Addiction Informed by DAT, SERT, and NET-Based Interactome Networks.**

- Authors: Feng H, Gao K, Chen D, Shen L, Robison AJ, Ellsworth E, Wei GW

- Source: PubMed

- PMID: 35294204

- Link: https://pubmed.ncbi.nlm.nih.gov/35294204/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**ASPIRER: a new computational approach for identifying non-classical secreted proteins based on deep learning.**

- Authors: Wang X, Li F, Xu J, Rong J, Webb GI, Ge Z, Li J, Song J

- Source: PubMed

- PMID: 35176756

- Link: https://pubmed.ncbi.nlm.nih.gov/35176756/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Learning spatial structures of proteins improves protein-protein interaction prediction.**

- Authors: Song B, Luo X, Luo X, Liu Y, Niu Z, Zeng X

- Source: PubMed

- PMID: 35018418

- Link: https://pubmed.ncbi.nlm.nih.gov/35018418/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Protein–protein interaction prediction with deep learning: A comprehensive review**

- Authors: Farzan Soleymani, Eric Paquet, Herna Lydia Viktor, Wojtek Michalowski, Davide Spinello

- Source: OpenAlex

- DOI: 10.1016/j.csbj.2022.08.070

- Link: https://doi.org/10.1016/j.csbj.2022.08.070

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Protein Function Analysis through Machine Learning**

- Authors: Chris Avery, John Patterson, Tyler J. Grear, Theodore D. Frater, Donald J. Jacobs

- Source: OpenAlex

- DOI: 10.3390/biom12091246

- Link: https://doi.org/10.3390/biom12091246

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Dowker complex based machine learning (DCML) models for protein-ligand binding affinity prediction**

- Authors: Xiang Liu, Huitao Feng, Jie Wu, Kelin Xia

- Source: OpenAlex

- DOI: 10.1371/journal.pcbi.1009943

- Link: https://doi.org/10.1371/journal.pcbi.1009943

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**XGboost-Ampy: Identification of AMPylation Protein Function Prediction Using Machine Learning**

- Authors: Zar Nawab Khan Swati, Ali Ghulam, Muhammad Sohail, Jawad Usman Arshed, Rahu Sikander, Muhammad Shahid Malik, Nauman Khan

- Source: OpenAlex

- DOI: 10.21015/vtcs.v10i2.1347

- Link: https://doi.org/10.21015/vtcs.v10i2.1347

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Prediction of Diabetes Empowered With Fused Machine Learning**

- Authors: Usama Ahmed, Ghassan F. Issa, Muhammad Adnan Khan, Shabib Aftab, Muhammad Farhan Khan, Raed A. Said, Taher M. Ghazal, Munir Ahmad

- Source: OpenAlex

- DOI: 10.1109/access.2022.3142097

- Link: https://doi.org/10.1109/access.2022.3142097

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Hom-Complex-Based Machine Learning (HCML) for the Prediction of Protein–Protein Binding Affinity Changes upon Mutation**

- Authors: Xiang Liu, Huitao Feng, Jie Wu, Kelin Xia

- Source: OpenAlex

- DOI: 10.1021/acs.jcim.2c00580

- Link: https://doi.org/10.1021/acs.jcim.2c00580

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

### 2021

**Interpreting machine learning models to investigate circadian regulation and facilitate exploration of clock function.**

- Authors: Gardiner LJ, Rusholme-Pilcher R, Colmer J, Rees H, Crescente JM, Carrieri AP, Duncan S, Pyzer-Knapp EO

- Source: PubMed

- PMID: 34353905

- Link: https://pubmed.ncbi.nlm.nih.gov/34353905/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Radiomic Detection of EGFR Mutations in NSCLC.**

- Authors: Rossi G, Barabino E, Fedeli A, Ficarra G, Coco S, Russo A, Adamo V, Buemi F

- Source: PubMed

- PMID: 33148663

- Link: https://pubmed.ncbi.nlm.nih.gov/33148663/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**DeepVF: a deep learning-based hybrid framework for identifying virulence factors using the stacking strategy.**

- Authors: Xie R, Li J, Wang J, Dai W, Leier A, Marquez-Lago TT, Akutsu T, Lithgow T

- Source: PubMed

- PMID: 32599617

- Link: https://pubmed.ncbi.nlm.nih.gov/32599617/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Towards the Interpretability of Machine Learning Predictions for Medical Applications Targeting Personalised Therapies: A Cancer Case Survey.**

- Authors: Banegas-Luna AJ, Peña-García J, Iftene A, Guadagni F, Ferroni P, Scarpato N, Zanzotto FM, Bueno-Crespo A

- Source: PubMed

- PMID: 33922356

- Link: https://pubmed.ncbi.nlm.nih.gov/33922356/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Multi-Target In Silico Prediction of Inhibitors for Mitogen-Activated Protein Kinase-Interacting Kinases.**

- Authors: Halder AK, Cordeiro MNDS

- Source: PubMed

- PMID: 34827668

- Link: https://pubmed.ncbi.nlm.nih.gov/34827668/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

**Structure-based protein function prediction using graph convolutional networks**

- Authors: Vladimir Gligorijevi■, P. Douglas Renfrew, Tomasz Ko■ció■ek, Julia Koehler Leman, Daniel Berenberg, Tommi Vatanen, Chris Chandler, Bryn C. Taylor

- Source: OpenAlex

- DOI: 10.1038/s41467-021-23303-9

- Link: https://doi.org/10.1038/s41467-021-23303-9

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

## Recommended Next Steps

1. Open the attached individual PDFs (or follow the DOI/URL links).

2. For each relevant paper, extract: main method, datasets used, evaluation metrics, key quantitative results, and stated limitations.

3. Record contradictions or gaps across papers.

4. Only after verification, move evidence into a formal manuscript.

## Scientific Rule

AI is an assistant, not a source. Never submit an AI-generated or placeholder claim without checking the original paper.

Generated on 2026-10-13 by Daily Bioinformatics Paper Writer.